



Hashcrack

hashcrack

Easy

Cryptography

picoCTF 2025

browser_webshell_solvable

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Description

A company stored a secret message on a server which got breached due to the admin using weakly hashed passwords. Can you gain access to the secret stored within the server?
Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.
Its current status is: **NOT_RUNNING**

Launch Instance

Hints ?

1

2

3

Understanding hashes is very crucial.
[Read more here.](#)

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Author: The Analyst: Hyposelenia

Challenge: hashcrack

Category: Cryptography

Date: 02/26/26



I. Objective

The objective of this challenge is to **identify different hash types and crack them by finding their original plaintext values**, demonstrating why weak or commonly used hashes are insecure.

II. Background

In cryptography, a **hash** is a one-way transformation that converts readable text (like a password) into a fixed-length string of characters. Hashing is commonly used to store passwords securely.

However, **not all hashes are equally secure**. Some older hashing algorithms can be easily cracked using online databases and tools, especially if the password is simple or commonly used.

This challenge introduces:

- Hash identification
- Hash cracking
- The importance of **strong hashing algorithms and strong passwords**

III. Tool Used

- **Kali Linux (Oracle VirtualBox)** – Used as the working environment
- **nc (netcat)** – Used to connect to the challenge server
- **Online Hash Identifier** – Used to determine the hash type (https://hashes.com/en/tools/hash_identifier)
- **Online Hash Decryptor** – Used to retrieve plaintext values (<https://10015.io/tools>)



IV. Methodology

1. Launched the challenge instance.

Its current status is: **NOT_RUNNING**

Launch Instance

2. Opened Kali Linux using Oracle VirtualBox.
3. Connected to the challenge server using the provided nc command.

```
File Actions Edit View Help
(kali@kali)-[~]
$ nc verbal-sleep.picoc.tf.net 56992
```

4. Observed the first hash displayed by the server.

```
File Actions Edit View Help
(kali@kali)-[~]
$ nc verbal-sleep.picoc.tf.net 56992
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: █
```

5. Copied the hash from the terminal using **CTRL + SHIFT + C**

(Note: Copying and pasting in the Linux terminal uses CTRL + SHIFT + C/V, not the usual CTRL + C/V.)

6. Pasted the hash into an online hash identifier to determine its type.



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hashes.com/en/tools/hash_identifier

Hashes.com Home FAQ Deposit to Escrow Purchase Credits API Tools Decrypt Hashes Escrow Support English

Identify hash types

Identify and detect unknown hashes using this tool. This page will tell you what type of hash a given string is. If you want to attempt to Decrypt them, click this link instead. [Decrypt Hashes](#)

Hashes (max. 25 separated by newline, format 'hash[:salt]')

482c811da5d5b4bc6d497ffa98491e38

☐ Include all possibilities (expert mode)

SUBMIT & IDENTIFY

7. Confirmed that the hash was **MD5**.

Hashes.com Home FAQ Deposit to Escrow Purchase Credits API Tools Decrypt Hashes Escrow Support English

Proceeded!
1 hashes were checked: 1 possibly identified 0 no identification

Pay professionals to decrypt your remaining lists
<https://hashes.com/en/escrow/view>

Possible identifications: [Decrypt Hashes](#)

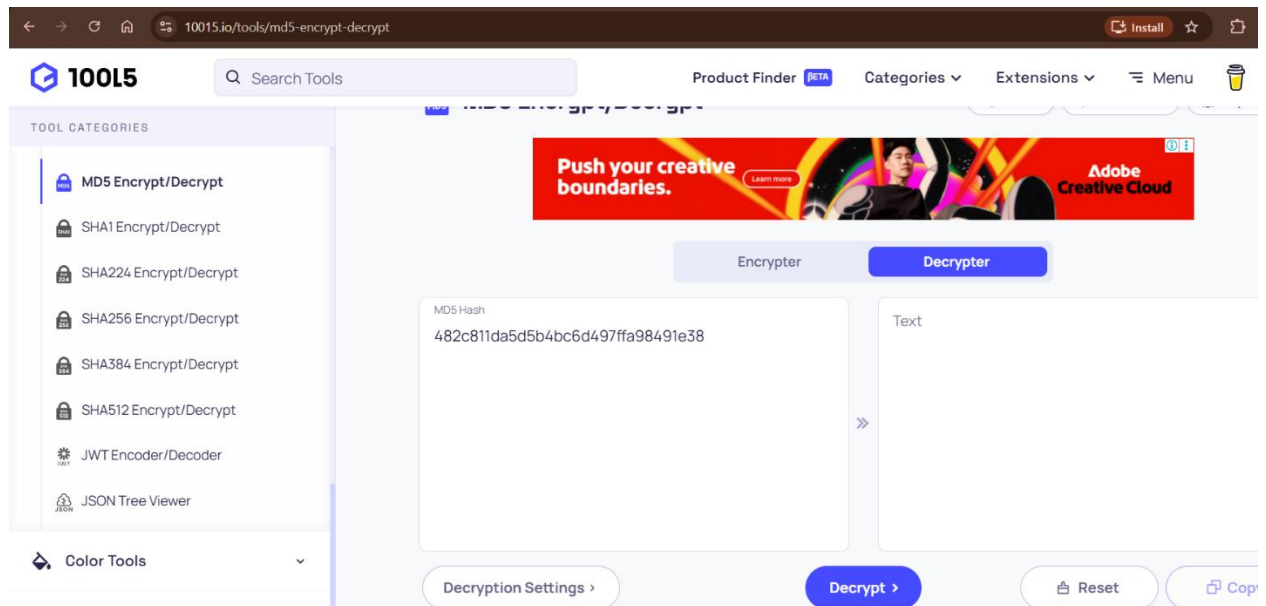
482c811da5d5b4bc6d497ffa98491e38 - Possible algorithms: MD5

SEARCH AGAIN

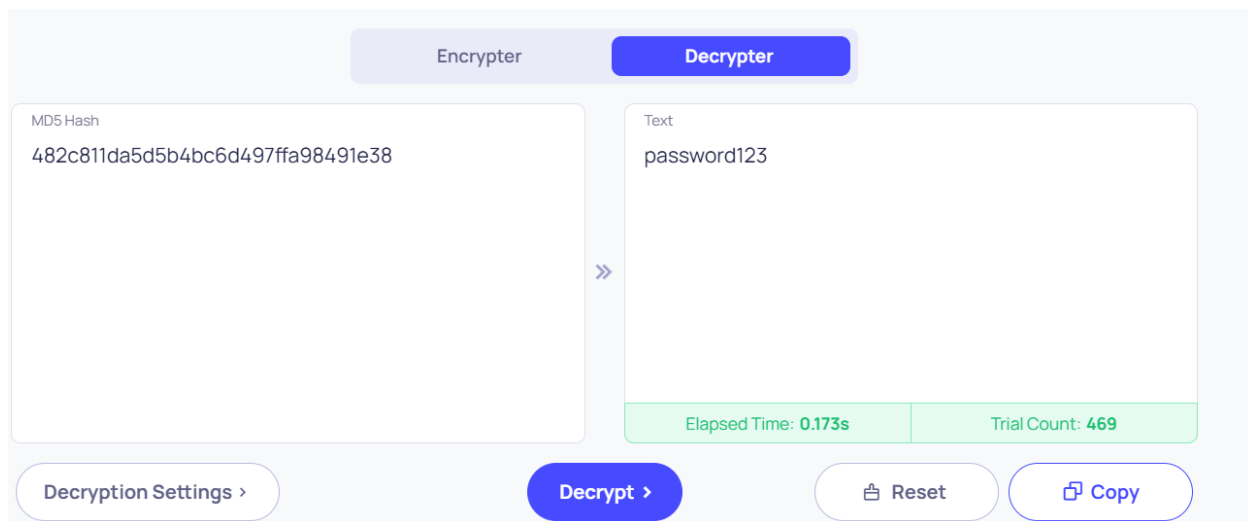
8. Opened the online hash decryptor and selected **MD5**.



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9. Pasted the hash and decrypted it to obtain the plaintext password.



10. Returned to the terminal and pasted the plaintext password using
CTRL + SHIFT + V.



```
nc verbal-sleep.picoctf.net 56992

Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: password123
Correct! You've cracked the MD5 hash with no secret found!

Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
Enter the password for the identified hash: █
```

11. Received a second hash from the server.

```
Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
Enter the password for the identified hash: █
```

12. Repeated the identification process and determined the second hash was **SHA-1**.

Proceeded!
1 hashes were checked: 1 possibly identified 0 no identification

Pay professionals to decrypt your remaining lists
<https://hashes.com/en/escrow/view>

✓ **Possible identifications:** [Decrypt Hashes](#)

b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3 - Possible algorithms: SHA1

[SEARCH AGAIN](#)

13. Used the online decryptor to crack the SHA-1 hash.



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TOOL CATEGORIES

- MD5 Encrypt/Decrypt
- SHA1 Encrypt/Decrypt**
- SHA224 Encrypt/Decrypt
- SHA256 Encrypt/Decrypt
- SHA384 Encrypt/Decrypt
- SHA512 Encrypt/Decrypt
- JWT Encoder/Decoder
- JSON Tree Viewer

Color Tools

Social Media Tools

SHA1 Encrypt/Decrypt

Share Add to Favs Report Bug

Encrypter Decrypter

SHA1 Hash

b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3

Text

letmein

Elapsed Time: 0.171s Trial Count: 192

Decryption Settings > Decrypt > Reset Copy

14. Entered the decrypted plaintext into the terminal.

```
Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
Enter the password for the identified hash: letmein
Correct! You've cracked the SHA-1 hash with no secret found!
```

15. Received a third hash from the server.

```
Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745
Enter the password for the identified hash: █
```

16. Identified the third hash as **SHA-256**.

Proceeded!
1 hashes were checked: 1 possibly identified 0 no identification

Pay professionals to decrypt your remaining lists
<https://hashes.com/en/escrow/view>

Possible identifications: [Decrypt Hashes](#)

916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745 - Possible algorithms: SHA256

SEARCH AGAIN



17. Used the online decryptor to recover the plaintext value.

10015 Search Tools Product Finder BETA Categories Extensions Menu

TOOL CATEGORIES

- MD5 Encrypt/Decrypt
- SHA1 Encrypt/Decrypt
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- SHA256 Encrypt/Decrypt**
- SHA384 Encrypt/Decrypt
- SHA512 Encrypt/Decrypt
- JWT Encoder/Decoder
- JSON Tree Viewer

Color Tools Social Media Tools

SHA256 Encrypt/Decrypt

SHA256 Hash

916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745

Text

qwerty098

Elapsed Time: 0.305s Trial Count: 63K

Decryption Settings Decrypt Reset Copy

18. Entered the final plaintext password into the terminal.

```
Correct! You've cracked the SHA-1 hash with no secret found.  
Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745  
Enter the password for the identified hash: qwerty098  
Correct! You've cracked the SHA-256 hash with a secret found.
```

19. Successfully retrieved the flag.

V. Result

picoCTF{UseStr0nG_h@shEs_&PaSswDs!_5b836723}

```
Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745  
Enter the password for the identified hash: qwerty098  
Correct! You've cracked the SHA-256 hash with a secret found.  
The flag is: picoCTF{UseStr0nG_h@shEs_&PaSswDs!_5b836723}
```

```
(kali@kali)~[~]
```




VI. Explanation

What is a hash?

Imagine you put a fruit (word) into a **blender**.

The blender:

- Always gives the **same smoothie** for the same fruit (word)
- Can't turn the smoothie back into the original fruit (word)

That smoothie is a **hash**.

What is MD5?

- MD5 is an **old hashing method**
- Produces a **128-bit hash**
- Written as **32 hexadecimal characters**
- Very fast — but **too weak**

MD5 is broken because attackers already know the hashes of common passwords.

What is SHA-1?

- More secure than MD5, but still outdated
- Produces a **160-bit hash**
- Vulnerable to collisions

SHA-1 is no longer recommended for security use.

What is SHA-256?

- Part of the SHA-2 family
- Produces a **256-bit hash**
- Written as **64 hexadecimal characters**
- Still considered secure today

Much harder to crack, especially with strong passwords.

Why cracking worked in this challenge

This challenge used:

- Weak or common passwords
- Hashes already stored in public databases

So instead of “breaking” the hash mathematically, the tools simply **looked up known results**.



Important cybersecurity lesson

Hashing is only as strong as:

- The algorithm used
- The password chosen

Even SHA-256 becomes weak if the password is simple.

Remember that I stated that a smoothie cannot be turned into its original fruit form. So how were we able to turn that hash into plaintext?

“If a smoothie can’t turn back into a fruit, why were we able to decrypt it?” *-maybe you asked*

A **hash** really **cannot be reversed**. So, we did not actually “decrypt” the hash. What we did was **recognize the smoothie**.

Imagine:

- **You blend banana + milk + sugar**
- **You get a banana smoothie**

You cannot unblend it back into a banana.

But... If someone already tasted that exact smoothie before, they can say:

“Oh! I know this flavor — it’s banana!”

That’s exactly what happened here.

What is a collision? (

A collision happens when:

Two different inputs produce the same hash

Example (not real):

hash("cat") = X

hash("dog") = X

That should never happen — but weak hashes allow it.

Why collisions are dangerous!

If attackers can cause collisions:



- They can fake passwords
- They can fake files
- They can break digital signatures

That's why:

- MD5 (✗ broken)
- SHA-1 (✗ broken)
- SHA-256 (✓ still trusted)

VII. Conclusion

This challenge highlights the dangers of weak hashing algorithms and poor password practices. By identifying and cracking MD5, SHA-1, and SHA-256 hashes, we learned how attackers exploit known databases and why modern systems must use strong hashing algorithms combined with complex passwords. Proper cryptographic practices are essential to maintaining secure systems.

— **The Analyst: Hyposelenia**